

Program-1: Build a Docker Container from a Custom Dockerfile

1. Files in the project

Program_1/

- |— app.py
- |— Dockerfile
- |— requirements.txt

2. Contents of each file

app.py

```
GNU nano 7.2 app.py
from flask import Flask

app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello, Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

requirements.txt

```
GNU nano 7.2 requirements.txt
Flask
```

Dockerfile

```
GNU nano 7.2 Dockerfile
# Use a lightweight Python base image
FROM python:3.9-slim

# Set working directory inside container
WORKDIR /app

# Copy and install dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Copy the application code
COPY . .

# Set Flask environment variables
ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

# Expose the Flask port
EXPOSE 5000

# Command to run Flask
CMD ["flask", "run"]
```

3. Build the Docker image

`docker build -t flask-docker-app .`

Once done, check your image:

`docker images`

4. Run the container

`docker run -d -p 5000:5000 --name flask_container flask-docker-app`

Check if it's running:

`docker ps`

```

1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program1$ nano app.py
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program1$ nano requirements.txt
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program1$ nano Dockerfile
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program1$ docker build -t flask-docker-app .
[+] Building 10.9s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 488B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd1e13cf1731b1b
=> [internal] load build context
=> => transferring context: 748B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:a1cf37813c3adb0b316d48624f141eec9d357421db2070c9ab0a069974791fd5
=> => naming to docker.io/library/flask-docker-app
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program1$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
flask-docker-app    latest             a1cf37813c3a       11 seconds ago     133MB
program4-app        latest             e1010a5e5baf       46 minutes ago     194MB
program-6           latest             c9d43c3a9ca5       3 hours ago        912MB
yashwanth142003/program-6   latest             c9d43c3a9ca5       3 hours ago        912MB
ghcr.io/yashwanth142003/program-6   latest             c9d43c3a9ca5       3 hours ago        912MB
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program1$ docker run -d -p 5000:5000 --name flask_container flask-docker-app
7a4506e16dd3ddb25fc3640a208e34132868f060fb27e7d9ca622ad694051413
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program1$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                               NAMES
7a4506e16dd3   flask-docker-app   "flask run"              17 seconds ago   Up 16 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   flask_container
1d3ec460d3b0   program4-app      "docker-entrypoint.s..." 47 minutes ago   Up 47 minutes   0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp   inspiring_haslett

```

Open in browser:

<http://localhost:5000>

